

Spectral Transfer

A research plan for label-efficient, geographically transferable Earth-observation representations

Train a compact multispectral encoder with self-supervision and investigate a specific invariance trade-off: does occasionally withholding a coherent spectral group improve transfer and missing-band robustness, or does it erase information needed for land-cover recognition?

Decision	Committed scope
Learning task	19-label land-cover classification on a bounded BigEarthNet v2 subset; this is multi-label classification, not single-label accuracy.
Own learning	ResNet-18 trained from random initialization using VICReg; compare RGB, full multispectral, and multispectral with asymmetric spectral-group dropout.
Evaluation	Six training-label budgets, fixed source-domain validation and test sets, and two held-out countries. Frozen probes supplement full fine-tuning.
External comparisons	ImageNet-pretrained ResNet-18, generic DINOv3 ViT-S/16, and OlmoEarth v1.1 Tiny. External pretraining is explicitly a confound.
Resources	One GPU; 8-12 weeks; target 50-70 GPU-hours and a hard planning cap of 80. Pilot measurements determine the final run budget.
Deliverable	Reproducible training code, split manifests, checkpoints, experiment ledger and a research-style technical report.

The intended contribution

The contribution is a controlled empirical study of spectral information, learned invariance and transfer under a fixed compute budget. Spectral dropout and VICReg are not claimed as new inventions. A careful negative result is acceptable; unsupported novelty or a promised performance gain is not.

How to use this plan

Follow the data and compute gates before launching the full matrix. All proposed sample counts and hyperparameters are starting commitments to audit, not measured dataset counts or optimal values. No training or dataset download has been performed for this plan.

Prepared for a solo B.Tech CS / AIML student applying for 2027 MSc entry. Research and tool availability were checked as of 13 September 2026. BigEarthNet v2, VICReg and released foundation models provide the technical starting points. ^{[1],[2],[10],[17],[18]}

1. Research question and hypotheses

Question: Under matched architecture and downstream label budgets, how do multispectral self-supervision and spectral-group invariance affect land-cover transfer to held-out European countries, relative to supervised learning and generic pretrained features?

ID	Testable hypothesis	Predeclared evidence
H1	Source-only multispectral SSL improves downstream training-label efficiency over the same supervised-from-scratch architecture.	M3 minus M0 mAP at 1%, 5%, 10%; report 25-100% too. Primary aggregate: area under the mAP curve over log training-label count.
H2	Additional spectral information helps beyond RGB when the SSL recipe, source data and training schedule are matched.	M3 minus M2; same U samples and crop seeds. Inspect class-wise changes, especially vegetation and water-related labels.
H3	Limited spectral-group dropout trades some spectral dependence for improved robustness or geographic transfer.	M4 minus M3; primary endpoints: country-averaged OOD mAP and withheld-band stress mAP at 10% labels. Report clean ID cost.
H4	Evaluation on held-out countries changes estimated performance and potentially model ranking relative to the source-domain test.	For each model, report ID mAP and each country's mAP, not just the gap. This is a compound-shift diagnosis, not a causal geography estimate.
H5	Transfer benefits are visible in encoder features, not solely in a flexible fine-tuning head.	Linear and k-NN probes before fine-tuning, paired nearest-neighbour analysis, effective rank and spectral sensitivity.

Interpretation rules

- For H1-H3, report paired differences and uncertainty; lack of positive evidence is a valid finding.
- An illustrative meaningful gain is 2 percentage points of OOD mAP with no more than 1 point clean-ID loss. These are predeclared practical thresholds, not expected results or significance cutoffs.
- Do not define success as beating OlmoEarth or DINOv3. Their scale, architecture, data and metadata differ from the controlled experiments.
- Use H4 to describe where methods fail. A geographic performance gap alone does not establish geographic leakage or show that a particular method learned causal features.

2. Dataset selection: alternatives

The data decision prioritizes verified geography, spectral fidelity, auditable labels and bounded storage. Dataset familiarity alone is insufficient. Published totals below describe the source releases, not the planned subset.

Dataset	Useful properties	Constraint and decision
BigEarthNet v2 [1],[2],[3]	549,488 S1/S2 pairs; S2 L2A; 19 multi-label classes; European geography; georeferenced bands; country and quality metadata.	PRIMARY. S2 archive is about 59 GiB. Use selected S2 patches only; omit S1 and dense reference-map downloads.
EuroSAT [4]	27,000 georeferenced 64 x 64 images, 13 S2 bands, 10 classes. All-band archive about 2.07 GB; distributed with MIT license metadata.	Engineering fallback. Small and easy to stage; weaker main study because restricted scenes and near-saturated random-split classification can obscure transfer.
SEN12MS [5]	180,662 georeferenced triplets, 13 optical bands, 256 x 256 patches, global ROIs; MODIS land-cover maps.	Useful SSL corpus, but coarse map supervision and larger I/O complicate this scope. The host lists 510 GB across roughly 542,000 files; stage selected ROIs if ever used.
SSL4EO-S12 / v1.1 [6],[7]	Global, multiseasonal, unlabeled data; original about 251k locations; 264 x 264 patches. v1.1 modernizes processing and distribution.	Excellent pretraining source, but v1.1's distribution totals about 2.26 TB. A second ingestion pipeline and overlap audit are unnecessary for the core.

Licensing and version discipline

BigEarthNet uses CDLA-Permissive-1.0. SSL4EO data and weights use CC-BY-4.0, while its repository code uses Apache-2.0. SEN12MS is distributed under CC-BY-4.0 on mediaTUM. EuroSAT's repository and the selected distribution expose MIT license metadata. Archive the exact release terms instead of inheriting the code license for unrelated upstream imagery. [1],[4],[5],[6],[7]

Label quality

BigEarthNet v2 refines processing and labels, but map-derived supervision is not pixel-perfect truth. Small objects, mixed patches and temporal mismatch between imagery and reference mapping can still produce errors. Treat outputs as recognition of benchmark land-cover labels, not validated environmental measurements. [2],[3]

3. Dataset selection: final strategy

Candidate	Assessment	Decision
So2Sat LCZ42 ^[8]	About 400k paired patches; 10 optical bands at 32 x 32; 17 urban/local-climate-zone classes. City/cultural holdouts and later geolocation releases are attractive. One original host release is 51.8 GB; size varies by version. CC-BY-4.0.	Strong alternative, but small spatial support and urban LCZ interpretation redirect the domain. Keep as a later external test, not part of the core.
FMoW / WILDS ^[9]	Well-defined temporal/geographic shifts and metadata; the common WILDS version uses RGB, 224 x 224 images and 62 classes. Licensing follows the selected upstream distribution.	Not primary: the convenient shift benchmark does not satisfy the multispectral requirement. Sentinel variants add a separate data/version problem.
GEO-Bench-2 ^[21]	Current successor to GEO-Bench; selectable datasets including benv2, richer modalities and coordinates for many datasets; licenses are dataset-specific.	A useful packaging option after verifying original IDs, full band semantics and geography. Do not assume a curated subset preserves all countries or labels needed here.
PANGAEA ^[22]	Diverse EO evaluation tasks, resolutions and sensor types; download size and license depend on the task.	Use its baseline discipline as methodological guidance. Running the full suite would dilute this project.

One dataset, two disjoint roles

Use BigEarthNet v2 S2 for both a source-only unlabeled pretraining pool U and a sample-disjoint downstream labeled pool D. U and D share source-domain training geography but contain no identical or overlapping footprints. Neither pool contains validation, ID-test or held-out-country patches.

This tests transfer across samples and label budgets within a known acquisition family, plus transfer to held-out countries. It does not establish cross-dataset, cross-sensor or global generalization. A second raw pretraining dataset is deliberately not needed to answer the question.

Why the domain earns its place

Land-cover structure depends on visible, near-infrared, red-edge and short-wave infrared measurements. Geography changes vegetation, land use, class prevalence and acquisition conditions. Those factors make spectral invariance a substantive modelling choice and give geographic transfer an interpretable setting.

Use the dataset's recommended clean-patch metadata. Do not mix v1 imagery with v2 labels by filename approximation; the official metadata contains an explicit v1 mapping if needed. ^{[1],[2]}

4. Data acquisition and subset contract

Partition	Target count	Source of samples
U: unlabeled SSL	20,000	Source training groups only; strip labels from the SSL loader.
D: labeled training reservoir	12,000	Different footprints in source training groups; all fractions drawn from this one fixed reservoir.
V: source validation	2,000	Held-out source groups; fixed for every method, fraction and seed.
I: source-domain test	3,000	Different held-out source groups; inaccessible to model selection.
T-FI: geographic test	4,000	Finland; all its eligible locations excluded from U, D and V.
T-PT: geographic test	4,000	Portugal; same exclusion. Report this country separately.
Total target	45,000	Counts are caps/targets pending metadata and spatial audit; never relax separation to hit a quota.

Acquisition sequence

- Fetch metadata.parquet and the release description first. Record URL, release ID, byte count and checksum. Inspect country, labels, split, patch_id and quality fields.
- Reserve Finland and Portugal before model experiments. Parse MGRS tile ID and timestamp from patch_id; derive footprints from GeoTIFF CRS/transform. Group repeated acquisitions of the same location.
- Build candidate manifests before staging pixels. Preserve all 12 available S2 bands so the OlmoEarth baseline can use its documented input. Core models select ten.
- Use a cloud CPU machine with about 120 GB temporary disk, or stream the Zstandard-compressed tar and retain only selected members. Streaming saves disk, not necessarily the roughly 59 GiB transfer. Never fully extract all bands for the entire release.
- Create sharded, sequentially readable arrays; preserve native-band metadata and valid masks. Keep a durable copy of the selected subset so notebook resets do not trigger repeated downloads.

Storage estimate and audit gate

45,000 x 120 x 120 x 12 x 2 bytes is about 15.6 GB (14.5 GiB) for a uniform uint16 or float16 cache before compression. Budget 25-35 GB for masks, manifests, caches and checkpoints. A float32 image cache doubles the image portion; do not keep it all in RAM.

Week 1 must confirm viable group counts, source support for most labels, and target image coverage. If the 12k reservoir is infeasible after exclusions, reduce counts and recompute fractions. If a reserved country lacks usable geographic replication, publish the limitation or a pre-training protocol amendment; never swap targets after seeing model scores.

5. Geographic split protocol

Use a custom split and name it explicitly, for example **BENv2-SpectralTransfer-v1**. BigEarthNet v2 already improves geographic separation; this project's country holdouts are an additional experiment, not a correction of a purely random official split. Results must not be compared directly to the official leaderboard. [2],[3]

Split algorithm

- Reserve all Finland and Portugal footprints, including repeated scenes and boundary-crossing footprints, as target-only candidates. Remove any target-country footprint from source pools.
- Within the remaining eight countries, group by MGRS cell without acquisition date. Assign whole groups approximately 70/15/15 to source training, V and I using split seed 1729; optimize source label coverage only within this geographic constraint.
- Audit cross-cell overlap using transformed footprints in a common metric CRS. Remove training footprints within a 2.4 km buffer of V, I or target footprints; remove V footprints near I and targets too. Check repeated acquisitions and exact/near duplicates.
- Sample U and D without overlapping footprints from source training groups, balancing geographic group contribution rather than letting one dense tile dominate. U may be near D because both belong to the training domain; it must remain distant from evaluation areas.
- If whole-cell allocation is too coarse, use fixed 12 km blocks with the same buffer for source V/I only. Document shared-scene effects. Country targets remain completely excluded. Freeze this choice before training.

What ID means here

I is a **source-domain, spatially blocked test**, not a perfectly IID random sample. It shares the source-country mixture while holding out locations. T-FI and T-PT add country-level shift. Compare per-country results and maps; do not compress everything into one pooled OOD number.

Support and independence

Aim for at least five spatial groups per evaluation partition and report the exact count; fewer groups imply weak geographic uncertainty estimates. Tabulate class counts, label co-occurrence, acquisition month, orbit and country. Use all source-supported classes for the predeclared metric and disclose target classes lacking source support.

Target images may be used for deterministic integrity checks and coverage maps, but not for normalization fitting, SSL, augmentation selection, checkpoint choice or hyperparameter tuning. Target labels are used only in the frozen final evaluation and its predeclared diagnostics.

6. Multispectral preprocessing

Group	Bands	Native resolution / role
Visible	B02, B03, B04	10 m; core visible information. RGB model receives B04, B03, B02 in that order.
Near infrared	B08	10 m; vegetation structure and reflectance.
Red-edge / narrow NIR	B05, B06, B07, B8A	20 m; vegetation-related spectral information.
Short-wave infrared	B11, B12	20 m; moisture and material information.
Baseline-only bands	B01, B09	60 m; retain for the 12-band OlmoEarth input, omit from the ten-band core.
Unavailable band	B10	Not a BigEarthNet L2A input. Never fabricate a cirrus channel as measured data.

Core order is [B02, B03, B04, B08, B05, B06, B07, B8A, B11, B12]. A native patch spans about 1.2 km. Align each band to the B02 120 x 120 grid using CRS/transform-aware bilinear interpolation; nearest-neighbour for validity masks. Upsampling a 20 m band does not create 10 m information. [\[2\]](#),[\[3\]](#),[\[23\]](#)

Radiometry and invalid observations

- Read GeoTIFF scale, offset, nodata and raster masks. Confirm release conventions on sample patches. Do not blindly apply current Copernicus processing offsets to this reprocessed historical archive.
- For own models, convert to reflectance using verified release scaling, then fit per-band mean/std on valid U pixels only. Use these same statistics for source and target data; no per-image min-max scaling.
- Reject a patch if more than 1% of pixels are invalid in any core band; this is an initial conservative rule to freeze after the audit. Fill remaining invalid pixels with the source-band mean after resampling and preserve a mask for QA.
- Do not treat every zero as nodata if the raster metadata says otherwise. Exclude official cloud/shadow/snow patches; acknowledge residual haze and label noise.
- Keep checkpoint-specific preprocessing separate for DINOv3, ImageNet and OlmoEarth. Log raw quantiles and adapter outputs to catch double division, saturation, incorrect band order or silently added channels.

Do not add temporal stacks, SAR, vegetation indices as model inputs, or a segmentation task in the core. They would create additional experimental axes.

7. Augmentations and spectral invariance

Shared spatial transforms

For SSL, sample a large square crop retaining 80-100% of patch area, resize to 120 x 120, and apply a shared 90-degree rotation and horizontal/vertical flips across all bands. Each view has its own crop; require crop intersection-over-union of at least 0.6 to reduce false positive pairs in mixed land-cover patches. Use identical random crop seeds across M2-M4.

For labeled training, retain the full footprint and use only flips/90-degree rotations. Aggressive crops can remove a minority class while preserving its scene-level label. Evaluation uses the full image without test-time augmentation.

Proposed treatment: asymmetric spectral-group dropout

In M4 only, with probability $p = 0.25$, replace one non-visible spectral group in one randomly chosen SSL view with the source-band mean (zero after standardization). Choose uniformly among {B08}, {B05,B06,B07,B8A}, and {B11,B12}. The other view remains complete. Keep all visible bands and never drop more than one group. M3 uses $p = 0$.

The architecture does not receive a dropout mask channel. It learns agreement between complete and partially observed views. This is a controlled augmentation/objective intervention, not a new sensor simulator. At downstream fine-tuning, M2-M4 receive the same clean-input augmentations; dropout is not continued, so the comparison isolates pretraining.

Why this is a research question

Red-edge and SWIR information may be essential for particular labels. Enforcing invariance to their absence can improve redundancy and robustness, or suppress useful signal. H3 explicitly tests both possibilities and reports the clean-accuracy cost.

Do not import RGB recipes blindly

- No hue shifts, grayscale conversion, solarization, arbitrary channel permutation or independent strong colour jitter.
- No random band dropping in RGB, and no mixing physically unrelated bands into pseudo-RGB for the core treatment.
- Use no radiometric jitter in the main training recipe. If studied later, use a small shared gain and report it as a sensitivity check; it is not a faithful atmospheric model.
- Do not call spectral-group dropout MAE or masked reconstruction: there is no reconstruction decoder or pixel prediction loss here.

8. Own model and learning objective

Use the standard torchvision ResNet-18 stem (7 x 7, stride 2, max-pooling), residual stages and global average pooling. Preserve the stem for ImageNet compatibility. RGB features have dimension 512; the multispectral version changes only conv1 from 3 to 10 input channels.

Component	Specification
Encoder	About 11.18M parameters for RGB and 11.20M for ten bands, excluding the classifier. Log actual counts from instantiated models.
SSL projector	512 -> 1024 -> 1024 -> 256 MLP; BatchNorm and ReLU after the first two layers, linear final layer. About 1.84M additional parameters.
Downstream head	Fresh 512 -> 19 linear layer with independent sigmoid outputs, approximately 9.7k parameters. Discard the SSL projector.
SSL initialization	Random initialization for M2, M3 and M4. Same U, steps, optimizer settings and per-seed initialization scheme.
Downstream adaptation	Train the full encoder and new head. Frozen features are secondary probes, not the sole learning achievement.

VICReg loss

```
z = g(f(view_1)); z' = g(f(view_2))
L = 25 * s(z, z') + 25 * [v(z)+v(z')] + [c(z)+c(z')]
s = mean over samples and dimensions of (z-z')^2
v(z) = mean_j max(0, 1 - sqrt(Var_batch(z_j) + 1e-4))
C(z) = (z-centered)^T (z-centered) / (B-1)
c(z) = sum_{i != j} C(z)[i,j]^2 / d
```

State this exact normalization in code and the report: both view penalties are summed, and covariance uses an unbiased denominator B-1. Keep loss reductions consistent when comparing runs. The variance term discourages constant features; covariance penalizes redundant dimensions; invariance aligns paired views. VICReg is an established method. ^{[10],[11]}

Stability checks

Compute variance/covariance terms in float32 even under mixed precision. Use a physical batch of 128 if possible, 64 as fallback, and a 256-dimensional projector output. Gradient accumulation does not enlarge the batch used to estimate covariance. Log each term, embedding standard deviations, gradient norms and effective rank; a low total loss alone does not demonstrate useful features.

Before long runs, test identical versus shuffled views, variance on constant features, gradient flow and overfitting a tiny labeled batch. Avoid claiming that moderate-batch VICReg is guaranteed to avoid collapse in this setting.

9. Why this SSL family

The project needs one interpretable objective and a small number of informative interventions. The decision below concerns solo feasibility and diagnostic value, not a universal ranking of SSL algorithms.

Family	Strength / limitation here	Role
SimCLR ^[12]	Simple contrastive objective; semantically similar land-cover patches can be false negatives, and large negative batches are expensive.	Considered; do not add a contrastive sweep.
MoCo ^[13]	Queue decouples negatives from physical batch size; momentum encoder and queue add state and debugging.	Good alternative if a separate objective study is later justified.
BYOL ^[14]	No explicit negatives; teacher, predictor and EMA schedules introduce additional mechanisms.	Not needed for the central spectral question.
DINO / iBOT ^{[15],[16]}	Strong teacher-student and patch-level representations; centering, temperatures, multi-crop or masked-token paths add complexity.	Use DINOv3 as an external baseline, not a from-scratch replica.
MAE / SatMAE ^[19]	Explicit reconstruction and spectral/temporal masking; masked ViTs can be efficient.	Credible alternative, but decoder capacity and low-level reconstruction would add confounds.
VICReg / Barlow Twins ^{[10],[20]}	Explicit representation regularization; covariance/correlation mechanisms can be inspected directly.	Choose VICReg. Keep Barlow Twins as background reading, not another required run grid.
EO teacher-student models ^{[18],[24]}	Modern multimodal representations and domain-aware objectives; released small variants exist.	Use a compact OlmoEarth checkpoint as context. Do not reproduce global pretraining.

What the student owns

Implement and verify the data contract, two-view multispectral transforms, VICReg objective, training loop, spectral-dropout intervention, geographic protocol and analyses. Reusing a ResNet implementation or released weights is appropriate; the intellectual work is the testable design and its evidence.

Novelty boundary

Before writing a paper, search specifically for spectral dropout, channel masking, sensor dropout and redundancy-reduction SSL in remote sensing. A familiar intervention can support a valuable reproducibility or empirical study. Do not rename it as a new algorithm to imply novelty.

10. Baseline registry and fairness

ID / purpose	Architecture and inputs	Learning / coverage
M0: supervised scratch	ResNet-18, ten bands, 120 px; ~11.20M encoder.	Train on labeled D fractions only. Controlled reference for H1.
M1: ImageNet supervised	ResNet-18, ten-band stem adaptation, 120 px; same architecture.	ImageNet initialization, then full fine-tuning. No EO SSL.
M2: own RGB SSL	ResNet-18, RGB, 120 px; ~11.18M.	VICReg on U, p=0, then full fine-tuning.
M3: own MS SSL control	ResNet-18, ten bands, 120 px; ~11.20M.	VICReg on U, p=0, then full fine-tuning.
M4: proposed treatment	Same as M3.	VICReg on U, p=0.25 spectral-group dropout, then clean full fine-tuning.
M5: generic SSL foundation	DINOv3 ViT-S/16, LVD web-pretrained, RGB; 21M; resize full field to 128 px.	Frozen probes at all fractions; full fine-tuning at 1%,10%,100%.
M6: EO foundation	OlmoEarth v1.1 Tiny; 12.5M encoder; 12 S2 bands; native 120 px.	Frozen probes at all fractions; full encoder fine-tuning at 1%,10%,100%; discard pretraining decoder.

M1 uses torchvision ResNet18_Weights.IMAGENET1K_V1. For each pretrained convolution, copy the RGB kernels to B04/B03/B02, fill other bands with the mean of the three kernels, and multiply all conv1 weights by 3/10. This is a declared heuristic, not an optimal multispectral initialization. Fine-tune all weights.

M1's ten-band input uses the same U-fitted normalization as M0/M3/M4. The extra ImageNet RGB control uses the checkpoint-compatible RGB mapping described here. Its comparison diagnoses the combined spectral-stem/preprocessing adaptation, not a perfectly isolated kernel effect.

Use the same physical images, downstream labels and tests for every method. M2-M4 isolate spectral/objective effects. M3-M0 compares the value of additional unlabeled learning, not equal total data exposure. M5/M6 are contextual performance references: architecture, pretraining data, band count, normalization and metadata prevent clean causal attribution.

At 10% labels add an ImageNet RGB ResNet-18 comparator (three runs). This checks whether M1's stem conversion created an artificially weak baseline. Report raw/trainable parameter counts, input channels, token count or feature dimension and trainable layers for every row. ^{[17],[18],[25]}

11. Pretrained-model compatibility

DINOv3: use the generic checkpoint

Select the official LVD/web-pretrained ViT-S/16, not a satellite-specialized checkpoint. Request weight access early; the release has its own license. If access is unavailable by week 2, substitute DINOv2 ViT-S/14 (about 22M), resize the full field to 126 px, and record the changed baseline. Do not silently mix DINOv2 and DINOv3 scores. ^{[17],[26]}

Create RGB reflectance from B04/B03/B02, apply one source-fitted bandwise 2nd-98th percentile display mapping to ^{[0],[1]}, then the checkpoint's documented ImageNet normalization. This is a domain input adapter, not native satellite calibration. Use no target percentiles or per-image histogram stretching. Use global mean patch features, excluding register tokens, consistently in probes and fine-tuning.

OlmoEarth: native optical inputs

Pin OLMOEARTH_V1_1_TINY, a published 2026 family with 12.5M encoder parameters. v1.2 is available but is not needed to answer the question; a stable version commitment improves reproducibility. Use the official Normalizer and verify its expected raw input units against examples/configuration. ^[18]

```
S2 order:
B02 B03 B04 B08 B05 B06 B07 B8A B11 B12 B01 B09
Sample: [batch, height, width, time=1, bands=12]
Use real acquisition date; month indexing follows loader.
Patch size 8; native 120px is divisible by 8; input_res=10m.
```

Retain and correctly resample B01/B09; do not invent them from RGB. Use the documented MaskedOlmoEarthSample validity convention, exclude invalid/padded tokens from pooling, and do not load reference land-cover maps as inputs. Pool encoder spatial tokens into one patch feature. The actual date is an extra input compared with ResNet; disclose this context advantage.

Why other checkpoints are not mandatory

Galileo is relevant masked multimodal research, but adding it duplicates the EO comparator role. Prithvi EO 2.0 uses 300M/600M models and six HLS bands with different native resolution; SatMAE adds another architecture/decoder path. SigLIP 2 introduces language supervision without a need for text here. ^{[19],[24],[27],[28]}

If the OlmoEarth loader cannot be made correct within two implementation days, use TorchGeo's SSL4EO-S12 ResNet-18 MoCo checkpoint as the declared fallback. Its 13-band L1C pretraining requires selecting matching ten kernels and recording L1C/L2A shift; never fill nonexistent B10 with fake observations. Archive model licenses and hashes. ^{[6],[25]}

12. Starting training configurations

All values are starting configurations. Tune only on source validation, within a small, logged budget; freeze the selected recipe before final tests. No listed value is an established optimum for this dataset subset.

Setting	Own SSL M2-M4	Downstream M0-M4	Foundation M5/M6
Optimizer	AdamW	AdamW	AdamW
Initial LR	3e-4	Scratch: 3e-4; pretrained/SSL encoder: 1e-4; head: 1e-3	Encoder: 1e-5; head: 1e-3
Weight decay	1e-4	1e-4; exclude biases/norm scales	0.05; exclude biases/norm scales
Physical batch	128; fallback 64	64; fallback 32	32; fallback 16
Schedule	5-epoch warmup, cosine to 1e-6	Warmup 5% of steps, cosine	Same form
Duration	60 epochs over U	30 epochs, minimum 300 optimizer steps	20 epochs, minimum 300 steps
Loss	VICReg with fixed reductions	BCEWithLogitsLoss, unweighted	Same BCE
Regularization	Two-view transforms; dropout only M4	Shared flips/90-degree rotations; no label mixing	Same geometric transforms
Stability	AMP; FP32 covariance; grad clip 1.0	AMP; clip 1.0	AMP; clip 1.0
Checkpoint	Final fixed-epoch state; save every 10 epochs	Best source-V mAP; ties choose earlier	Best source-V mAP

Preserve standard BatchNorm during training; use evaluation mode for inference and frozen probes. Do not update BatchNorm on V//targets. Save normalization buffers, optimizer, scheduler, scaler and RNG states for resumable runs.

Bounded tuning

On the 10% source-label setting, try at most three learning-rate multipliers {0.3,1,3} per trainable method at one pilot seed and half duration. For SSL, test at most two learning rates with a short U pilot and a source-V linear probe. Keep $p=0.25$ and the loss coefficients fixed; no joint sweep.

Extend duration only if training and source-V curves show undertraining, apply the rule across affected comparators, and debit the budget. Foundation models can differ in optimization needs: report their smaller anchor grid and tuning budget rather than describing them as compute-matched.

A fixed large V set supplies labels for recipe selection. This is disclosed separately from the downstream training fractions; the study is not an end-to-end annotation-budget benchmark.

13. Label-efficiency protocol

Fraction of D	Training patches if D=12,000	Total with fixed V=2,000
1%	120	2,120
5%	600	2,600
10%	1,200	3,200
25%	3,000	5,000
50%	6,000	8,000
100%	12,000	14,000

Fixed and nested

- Build D, V, I and the two country tests once. Only the labeled subset of D changes. U stays identical across all fractions and methods.
- For seed bundles 17, 29 and 43, create nested D subsets with iterative multi-label stratification, respecting geographic sampling quotas. Reuse the exact selected patch IDs across every method.
- Use labels only to construct the experimental subsets, then release the manifests. State that label-aware benchmark subsampling is not an operational active-learning algorithm.
- Record positive counts for every class and fraction. Never force a rare class into a subset by duplicating images, and never resample until results look favorable.
- Reset the downstream head and optimizer for each fraction. Never warm-start 5% from 1% or reuse a model that has already seen 100% of D labels.
- The three bundles pair initialization, training RNG and subset realization. Their variance combines these effects; it does not separately estimate each source of variability.

What a low-label claim may say

Say 'using 10% of downstream training labels with a fixed 2,000-label validation set and 20,000 unlabeled pretraining patches.' Do not say 'using only 10% of all labels' or '90% lower annotation cost.' The labels used to stratify and develop the protocol are another experimental-design resource.

Rare labels and tiny budgets

Keep 1% as an explicitly high-variance stress point if several classes are absent. Define the source-supported reporting class set using full D before training, not by favorable target performance. If D is reduced below 8,000, replace percentages with declared counts and omit a tiny 1% point when it cannot support interpretable estimates.

Summarize curves using trapezoidal area over $\log_{10}(\text{training-label count})$, normalized by the log-count range. Show the individual fractions as well; do not infer a label-saving factor from non-overlapping or unstable curves.

14. Metrics and statistical analysis

Primary metrics

This is a multi-label task. Use macro average precision (mAP: average of per-class AP) as the principal threshold-free metric. Report source I, Finland and Portugal separately. OOD summary is the equal-weight average of the two country mAP values, not a patch-count-weighted pool.

Metric	Specification and limit
Class support	Predeclare C_{source} from full D : at least 50 positives and 50 negatives. Report all 19 classes individually. AP is undefined for a test class with no positives; mark NA, do not set it to zero.
Comparable ID/OOD gap	For the gap only, use the common supported classes with positive/negative examples in I and both countries. Derive this diagnostic once from final support counts; show which classes were excluded.
Thresholded performance	Macro F1, micro F1 and per-class recall at 0.5. Also select one global threshold maximizing source-V micro F1 on $\{0.05, 0.10, \dots, 0.95\}$; break ties toward 0.5, then lower. Freeze it for every test domain of that run.
Calibration	Mean binary Brier score across label decisions and macro classwise ECE with 15 equal-width bins. Report sample/support counts; pooled ECE can hide rare-class failure.
AUROC	Optional secondary classwise statistic only when positives and negatives exist. AP is usually more informative under severe imbalance.
Training efficiency	Actual GPU-hours, peak VRAM, examples/second, epochs/steps, and image exposures; include pretraining, adaptation and tuning.

Uncertainty and comparisons

Save per-patch logits. Report mean and standard deviation across three bundles. For paired model differences, bootstrap geographic groups (MGRS cells; fixed spatial blocks as a sensitivity analysis), resampling the same groups for both methods. Use 1,000 replicates, and recompute AP from the resampled predictions. Show each seed's estimate plus the seed-averaged paired difference.

Few geographic groups make bootstrap intervals unreliable; publish group counts and interpret intervals descriptively. Three seeds are not enough for strong normality assumptions or broad significance claims. Treat H1/H3 as primary endpoints and classwise/robustness analyses as exploratory; do not mine dozens of per-class p-values.

Acquisition and prevalence confounds

Describe quarter, orbit, sensor and class-prevalence differences. Add a predeclared common-quarter/per-class diagnostic when support permits. AP itself depends on prevalence, so a lower OOD mAP cannot be attributed solely to covariate shift. Do not reweight or relabel target data to manufacture a gain.

15. Experiment matrix and run ledger

Stage	Exact experiments	Run count / hypothesis
P0: pilots	Data QA, tiny overfit, two short SSL LR pilots, bounded downstream LR checks; no final tests.	Budgeted separately, maximum 6 GPU-hours.
P1: SSL	M2 RGB p=0; M3 MS p=0; M4 MS p=.25. Each trained on U with three bundles.	3 x 3 = 9 SSL runs; H1-H3/H5.
P2: controlled curves	M0-M4 x six label fractions x three bundles; evaluate each trained model on I and both countries.	5 x 6 x 3 = 90 downstream runs; H1-H4.
P3: foundation context	M5/M6 x {1%,10%,100%} x three bundles, full fine-tuning.	2 x 3 x 3 = 18 downstream runs; H1/H4 context.
P4: representation probes	Frozen M1-M6 features; six label fractions; three bundles. Train cheap linear heads on cached features; k-NN at 10%.	108 small probe fits; H5. Includes repeated cached external encoders; not 108 GPU trainings.
P5: one extra control	ImageNet RGB ResNet-18 at 10%, three bundles, same labeled images as M1.	3 downstream runs; H2 / stem-adapter check.
P6: diagnostics	M3/M4 at 10%: held-out-band stress, spectral sensitivity, CKA and nearest neighbours. Reuse existing checkpoints.	No additional model training; H3/H5.

Mandatory training ledger: 9 SSL runs plus 111 downstream training runs, with small datasets and short schedules. A run is not automatically a multi-hour job. The GPU budget is checked using measured per-step throughput before approval of the full queue.

Budget-first optional work

Only after the core fits the cap: (a) one 10k-U M3 pretraining run and 10% fine-tune to study unlabeled scale, or (b) one p=.5 M4 run and 10% fine-tune to study over-invariance. These are one-seed exploratory results, not added to the main three-seed claim. Choose at most one.

Random-split diagnostic

A patch-random source-only split may be illustrated as a separate optional diagnostic. If run, compare M1 under random and blocked source splits at the same training count and label/quarter mix; keep country tests out. Differences remain confounded by sample selection. Do not replace the main blocked protocol or portray random splitting as the official v2 protocol.

Avoid a Cartesian product over objective family, backbone, band set, masking probability, initialization, learning rate and dataset. Every new job must reference a hypothesis, comparator, expected information and remaining GPU budget.

16. Representation and robustness analysis

Analysis	Method	What it can establish
Linear probe	Freeze encoder in eval mode; standardize features using labeled training subset only; train 19-output logistic head with BCE and source-V-selected L2 regularization.	Whether useful label structure is accessible before encoder fine-tuning. Probe features, not projector outputs.
Multi-label k-NN	L2-normalize features; cosine neighbours among D_fraction only; weighted label averages. Choose k from {5,20,50} on V and fix it.	Local class structure without a learned deep head. Exclude self/duplicate matches.
Nearest neighbours	For 50 fixed ID and 50 fixed OOD queries, inspect top-5 source neighbours; summarize label-set Jaccard and distance/location patterns.	Concrete successes and shortcuts; manual examples illustrate, rather than replace, aggregate evidence.
CKA / effective rank	Use the same fixed 1,000 source-V patches across models. Compute centered linear CKA and covariance eigenvalue effective rank on 512-D encoder features.	Representation similarity and collapse/redundancy diagnostics. High CKA or rank is not proof of semantic quality.
Band sensitivity	Replace one spectral group with source mean; measure feature cosine change and AP loss for M3/M4 using the same images.	Dependence on information and synthetic missing-band tolerance. Mean replacement itself is a distribution shift.
Error slices	Per-class AP/recall, mixed-label count, quarter, country and source-neighbour distance; predeclare bins.	Which data conditions explain errors; causal interpretation remains limited.

Small, fixed stress suite

At the 10% label point, test clean images, missing B08, missing red-edge group, missing SWIR group, and a mild common reflectance gain of 0.9 or 1.1 before standardization. Do not retrain or tune thresholds for these conditions. Report absolute performance and degradation. No claim of cloud, atmospheric or cross-sensor robustness follows from these artificial perturbations.

Visualizations should answer a question: plot band-removal cost against class AP change, or overlay nearest-neighbour geography on retrieval errors. A UMAP/t-SNE figure, if included, uses fixed examples and parameters and carries no standalone evidence claim.

For M4, lower band sensitivity with lower clean AP may indicate over-invariance, not success. Compare the trade-off directly against M3. Do not equate confidence with calibrated uncertainty or OOD detection.

17. Compute model and stop rules

The following are planning estimates for a single 16-24 GB NVIDIA GPU, not benchmark measurements. A100/L4/T4 runtime can differ substantially. CPU I/O and notebook interruption may dominate. No paid instance is launched as part of this plan.

Work	Planning assumption	GPU-hours
Pilots and tuning	Short checks; early stop unpromising configurations using V only.	4-6
9 SSL runs	U=20k, 60 epochs. 9 x 1.2M paired examples / 200 paired examples/s = 15 h raw; add 40-60% overhead.	21-25
Controlled fine-tuning + RGB control	About 10.3M image exposures before step floors; roughly 350 images/s plus validation and small-run overhead.	10-14
DINOv3 / OlmoEarth anchors	~1.6M image exposures before minimum-step floors; assume 60-100 images/s.	8-12
Cached probes, inference, analyses	Batch feature extraction; CPU/small GPU linear heads; three test domains.	4-6
Recovery allowance	Preemptions, corrupt cache repair, limited justified reruns.	3-7
Expected total	Re-estimate after the warm-throughput pilot; never present this as measured cost.	50-70

Pilot-based calculation

For each run:

```
hours = planned_optimizer_steps * measured_seconds_per_step / 3600  
        + validation/checkpoint time
```

```
Total = sum(all runs) + tuning + extraction + recovery
```

Measure 50 warmup steps plus 200 timed steps on the staged data, with real augmentations and validation. For SSL, throughput is paired examples/second, already including both views. Do not divide by two again. Log physical batch and peak VRAM.

Hard cap and fallback order

- At projected total above 80 GPU-hours, first remove optional work, then make M5/M6 frozen-probe-only contextual baselines. Actual model development remains in M2-M4.
- If still too slow, keep three seeds but reduce U to 10k and SSL to 40 epochs, use 96px ResNet inputs, and retain label fractions {1%,10%,100%}. Recompute every paired control at the same size.
- Minimum viable study still includes M0/M1, RGB and multispectral SSL, dropout ablation, both geographic tests, and the two pretrained frozen references. Target 25-40 GPU-hours after a new pilot.
- If physical batch is below 64, reduce projector width/dimension and test stability. Accumulation is not a substitute for the covariance batch.

Cash estimate is H x quoted GPU hourly rate plus storage/egress. For illustration only, 60 hours at an assumed \$0.50-\$1.00/hour is \$30-\$60 before extras; these are not verified vendor quotes. Free Colab/Kaggle quotas and GPU allocation are not guaranteed.

18. Twelve-week execution roadmap

Week	Work package	Exit evidence
1	Read dataset and objective papers; inspect metadata; license/version ledger; reserve countries; audit candidate geography.	Frozen draft split manifest, map, label/support table, storage plan; data gate passes.
2	Build raster pipeline, shards, adapters and metric code; tiny overfit; measure GPU throughput; obtain checkpoints.	Valid data batches; no split overlap; reproducible smoke run; measured compute forecast.
3	Implement VICReg and matched view generator; verify covariance gradients and collapse diagnostics; short source-only pilots.	Finite stable losses, feature variance/rank logs and selected initial schedule.
4	Complete M2-M4 SSL pretraining for all three bundles; save complete run state.	Nine encoder checkpoints or documented budget fallback before the large matrix.
5	Run M0/M1/M2-M4 at 1%,10%,100%; begin probes; confirm no obvious comparator undertraining.	Primary anchor results on V only; all recipes frozen; test labels still sealed.
6	Complete 5%,25%,50% controlled curves; finish three bundles and ImageNet RGB control.	90+3 controlled downstream states; label manifest and timing checks complete.
7	Foundation baseline adapters/probes and full-tuning anchors if budget permits.	Two credible external comparator records, including preprocessing and contamination caveats.
8	Freeze analysis commit and checkpoint list; run final I/T evaluations; bootstrap and aggregate.	Locked prediction files, metrics and uncertainty tables; hypotheses answered honestly.
9	Representation analysis, stress tests, nearest-neighbour/error inspection.	Explanations connected to hypotheses, not decorative visualizations.
10	Write technical report, reproducibility run on fresh environment, repository documentation.	Mentor-reviewable report, rerunnable commands and provenance.
11-12	Reserved for mentor feedback, one justified extension or failure recovery; final résumé wording.	Complete artifact package; no unsupported novelty or performance claims.

Eight-week route

Combine weeks 1-2 and 3-4, use the minimum viable matrix if the throughput gate requires it, and merge analysis/documentation into weeks 7-8. Do not compress the schedule by eliminating geographic audits or seed replication.

Weekly review

Record completed jobs, compute used, unresolved defects and the next hypothesis to resolve. After final test inspection, further methodological changes require a versioned exploratory analysis or new untouched holdout; the same test must not become validation.

19. Repository and implementation blueprint

```
spectral-transfer/  
  configs/          # data, SSL, downstream, probes, run matrix  
  src/data/         # metadata, footprints, band adapters, shards  
  src/models/       # ResNet, projector, pretrained adapters  
  src/losses/       # VICReg with explicit reductions  
  src/train/        # SSL, fine-tune, linear probe, resume logic  
  src/eval/         # AP/F1, calibration, grouped bootstrap  
  src/analysis/     # CKA, spectral stress, nearest neighbours  
  scripts/         # audit, stage, train, evaluate, aggregate  
  manifests/       # versioned patch IDs, groups, label subsets  
  tests/           # scientific data/metric invariants  
  reports/         # figures, tables, technical report  
  README.md  DATA.md  MODEL_CARD.md  EXPERIMENTS.md
```

Configuration contract

Every job records model_id, checkpoint hash, input band order, normalization hash, data/split/label-subset hashes, SSL seed, downstream seed, hypothesis, optimizer, steps, physical batch, hardware and git commit. A central JSONL/CSV ledger records status, metrics and GPU time. Do not track raw private credentials, full data archives or giant checkpoints in Git.

Tests that protect scientific validity

- Assert disjoint patch IDs, footprints and buffered evaluation regions; forbid target-country samples in training/SSL loaders.
- Test band-order mapping and synthetic resampling, invalid-pixel handling, consistent RGB extraction, and checkpoint input shapes.
- Compare AP/F1 implementations with a small hand-checkable multi-label example; verify NA behavior and target-threshold isolation.
- Verify that frozen probes do not update encoder parameters or BatchNorm; resumed training restores optimizer/scheduler/RNG state.
- Check nested labels, fraction counts and no head/checkpoint carryover from larger label budgets.

Proposed command interface

```
python -m scripts.audit_data --config configs/data.yaml  
python -m scripts.train_ssl --config configs/ssl/m4.yaml --seed 17  
python -m scripts.finetune --model M4 --fraction 0.10 --seed 17  
python -m scripts.evaluate --run-id <id> --split test_finland  
python -m scripts.aggregate --ledger runs.jsonl
```

These commands specify the intended repository interface; no implementation is claimed to exist yet. Prefer straightforward PyTorch, torchvision, rasterio, NumPy, pandas/pyarrow, scikit-learn and a lightweight config system. Pin compatible versions only after week-2 smoke tests; do not stack frameworks for appearance.

20. Outcomes, risks and academic presentation

Risk	Response
SSL fails or collapses	Debug loss reductions, physical batch, normalization and view overlap. Report the failed configuration; do not conceal it behind a frozen-encoder result.
Spectral dropout hurts	Treat it as evidence of an invariance/information trade-off. Retain M3 as the strongest own model while reporting M4 honestly.
Foundation models dominate	Report their head start and pretraining uncertainty. A controlled small-model explanation remains valuable even without the best absolute score.
Geographic class support is poor	Publish support tables and restrict only predeclared metrics. Do not force cross-country comparability by removing inconvenient labels after scoring.
Unknown pretrained overlap	Label external models as geographically held out from this project's adaptation, not necessarily unseen during foundation pretraining. Only own random-init SSL has auditable source-only exposure.
Few country/tile groups	Limit the claim to these held-out regions. Wider continental/general domain claims require additional independent geography.
GPU or storage disruption	Use resumable jobs, durable shards and the preset smaller matrix. Preserve scientific controls before adding features.

Completion standard

Deliver the full run ledger including failures; one label-efficiency figure with source/OOD panels; per-country and per-class tables; a spectral-dropout ablation; representation evidence; compute and contamination disclosures; reproducible split manifests; and a technical report with limitations. No UI or deployment is required.

Résumé wording after results exist

Use measured placeholders only after validation: 'Trained a [parameter-count] multispectral encoder with VICReg on [N] source-region patches; evaluated label efficiency across [budgets] and held-out [regions], observing [measured effect and uncertainty].' A second bullet can describe spectral ablations and reproducible geographic evaluation. Do not put hypothetical gains or 'novel foundation model' on the résumé.

Paper extension gate

A paper is optional. Proceed only if a result survives matched controls, repeated seeds, a broader literature check and mentor review. The most useful extension would be one independent geographic split or dataset, or a mechanistic explanation of when spectral invariance helps. A workshop/reproducibility study may be appropriate; no venue outcome is promised.

Portfolio role: this demonstrates representation objectives and learning from real multispectral observations. ChronoPDE remains the continuous-dynamics project; FinSight remains the retrieval/system project. Finish the experimental evidence rather than adding another application layer.

References

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BigEarthNet v2.0 dataset and license

Current release; accessed 13 Sep 2026. [Primary source](#)

2. BigEarthNet authors

Description of BigEarthNet v2.0

Release documentation. [Primary source](#)

3. Clasen et al.

reBEN: Refined BigEarthNet Dataset for Remote Sensing Image Analysis

IGARSS 2025; arXiv revised May 2025. [Primary source](#)

4. Helber et al. / TorchGeo

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Original dataset and maintained distribution. [Primary source](#) | [Companion source](#)

5. Schmitt & Hughes / TUM

SEN12MS dataset

6 June 2019; DOI 10.14459/2019mp1474000. [Primary source](#)

6. Wang et al.

SSL4EO-S12: data, checkpoints and licensing

IEEE GRSM 2023; official repository. [Primary source](#)

7. Blumenstiel et al.

SSL4EO-S12 v1.1: updated dataset

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8. Zhu et al. / TUM

So2Sat LCZ42: versions, geolocation and band structure

GRSM 2020; maintained repository. [Primary source](#) | [Companion source](#)

9. Koh et al. / WILDS

FMoW domain-shift benchmark

WILDS dataset documentation. [Primary source](#)

10. Bardes, Ponce & LeCun

VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning

ICLR 2022. [Primary source](#)

References (continued)

11. Meta / facebookresearch

VICReg official PyTorch implementation

Archived reference code; MIT license. [Primary source](#)

12. Chen et al.

A Simple Framework for Contrastive Learning of Visual Representations

ICML 2020 (SimCLR). [Primary source](#)

13. He et al.

Momentum Contrast for Unsupervised Visual Representation Learning

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14. Grill et al.

Bootstrap Your Own Latent

NeurIPS 2020 (BYOL). [Primary source](#)

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Emerging Properties in Self-Supervised Vision Transformers

ICCV 2021 (DINO). [Primary source](#)

16. Zhou et al.

iBOT: Image BERT Pre-Training with Online Tokenizer

ICLR 2022. [Primary source](#)

17. Simeoni et al. / Meta

DINOv3: official models, input recipes and license

2025 technical report; repository checked Sep 2026. [Primary source](#) | [Companion source](#)

18. Ai2

OlmoEarth minimal implementation: versions, model sizes, input contract and normalization

v1.1/v1.2 documentation, accessed Sep 2026. [Primary source](#) | [Companion source](#)

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References (continued)

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GEO-Bench-2: data and evaluation framework

2025-2026 benchmark release. [Primary source](#) | [Companion source](#)

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PANGAEA: A Global and Inclusive Benchmark for Geospatial Foundation Models

Initially Dec 2024; benchmark source. [Primary source](#)

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Official technical documentation, accessed Sep 2026. [Primary source](#) | [Companion source](#)

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Official checkpoint metadata. [Primary source](#)

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SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features

February 2025 technical report. [Primary source](#)

Evidence boundary. Published dataset/model facts are cited. Sample allocations, interventions, hyperparameters, practical effect thresholds and compute estimates are the proposed protocol. They have not been validated by running the experiments. This plan makes no claim of a novel algorithm or an achieved performance improvement.